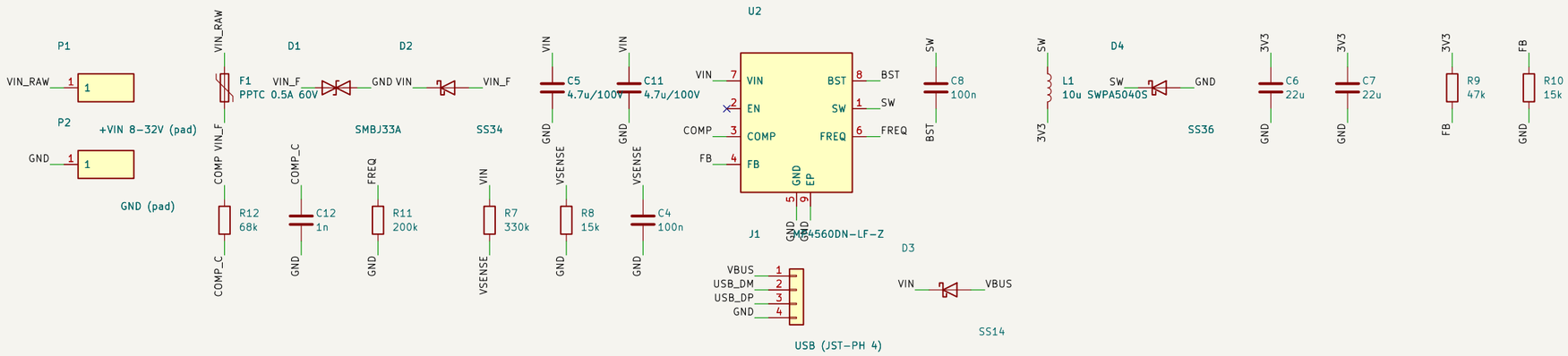
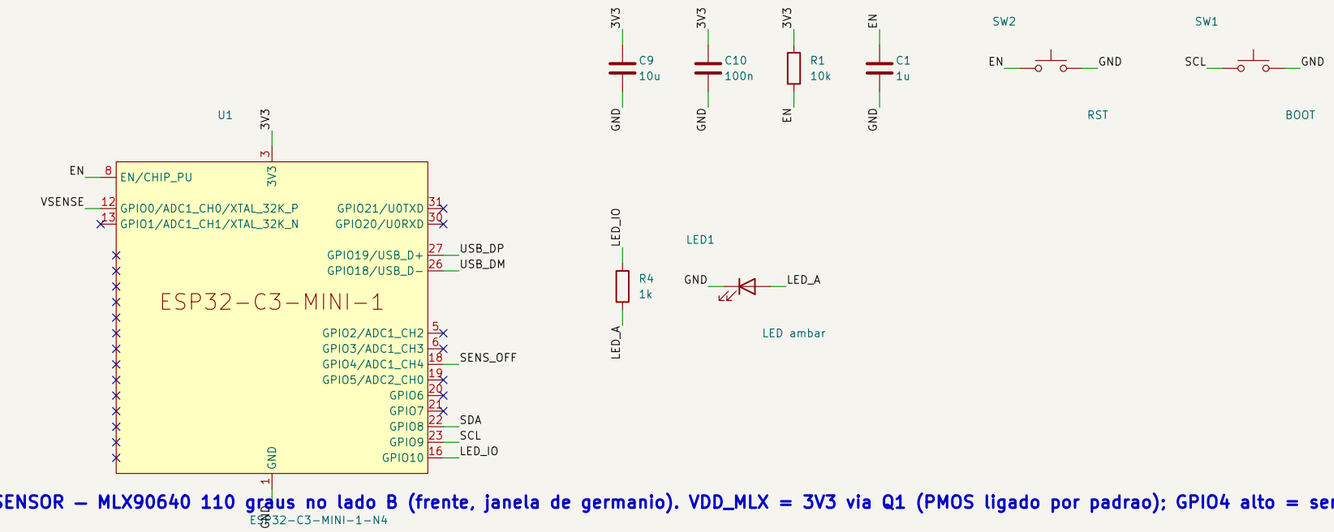
SAFEBOAT TERMICA v2 — camera termica MLX90640 110 + ESP32-C3, 8-32 V do barco, 4 camadas · esquematico p/ revisao

Redes por label (mesma netlist pino a pino de make_board.py). Placa 40 x 40 mm, 4 camadas (In1 GND, In2 3V3); sensor no lado B (frente), resto no lado F (re). Montagem completa JLCPCB (2 lados).

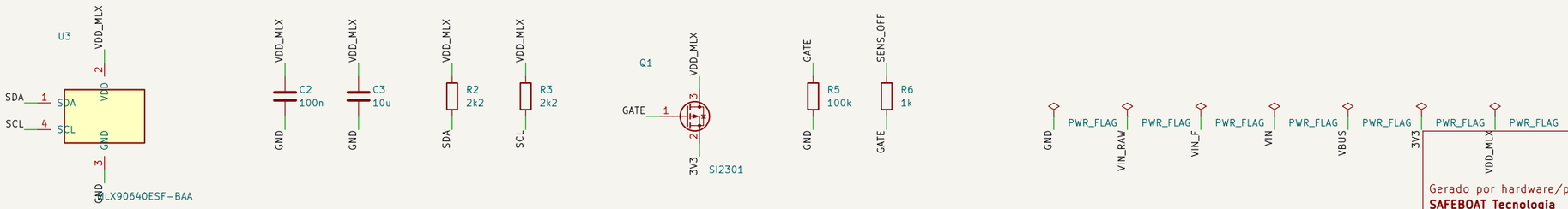
1 · ALIMENTACAO — +8-32 V (pads P1/P2, fio soldado) -> F1 PPTC 0,5 A -> D1 TVS SMBJ33A -> D2 SS34 (inversao) -> VIN -> U2 MP4560 (500 kHz, EN flutuante = liga) -> L1 10 uH / D4 SS36 -> 3V3 (FB 47k/15k) · USB J1 -> D3 SS14 ->



2 · MCU — ESP32-C3-MINI-1: GPIO8 SDA · GPIO9 SCL (+BOOT) · GPIO10 LED · GPIO4 SENS_OFF · GPIO0 VSENSE · GPIO18/19 USB · EN com R1/C1 + RST



3 · SENSOR — MLX90640 110 graus no lado B (frente, janela de germanio). VDD_MLX = 3V3 via Q1 (PMOS ligado por padrao); GPIO4 alto = sensor desligado. Pull-ups I2C p/ VDD_MLX.



Gerado por hardware/pcb/gen_schematic.py — netlist idêntica a make_board.py (PCB)

SAFEBOAT Tecnologia

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